

Statement of the Identification Problem

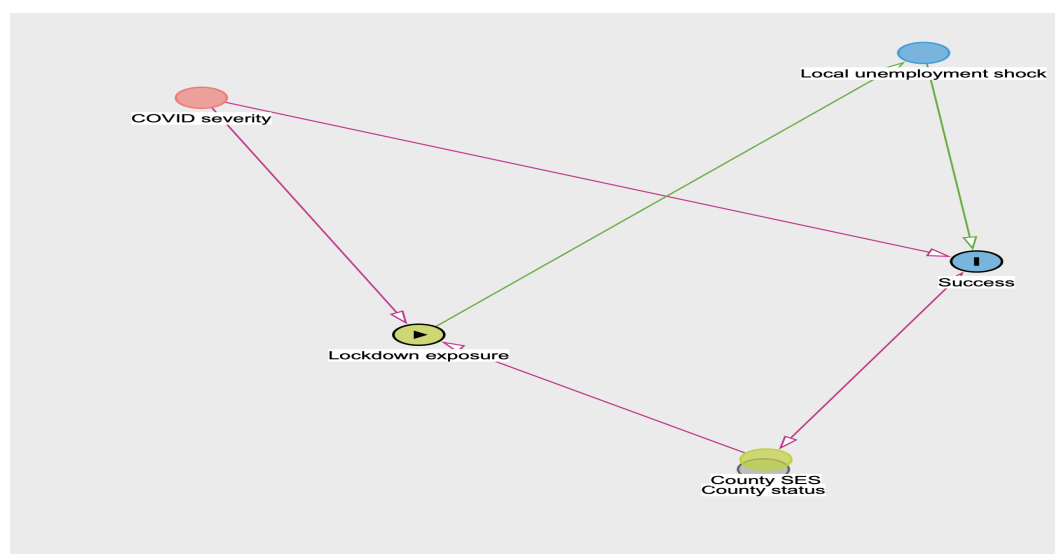
The central causal question of this study is: what is the effect of COVID-19 lockdown intensity on crowdfunding project performance? Answering this question is complicated by a fundamental identification challenge: counties that experienced more severe lockdowns may differ systematically from those that experienced milder restrictions in ways that independently affect crowdfunding outcomes.

Specifically, three endogeneity concerns must be addressed:

- Selection bias: Counties with high COVID-19 case rates (triggering stricter lockdowns) may differ in pre-existing economic conditions, internet infrastructure, or entrepreneurial culture, all of which shape crowdfunding success rates.
- Omitted variable bias: Unobserved county-level characteristics, such as local risk aversion, social capital, or income inequality, may simultaneously predict both lockdown severity and campaign performance.
- Reverse causality: While less plausible here, stronger local economic activity might influence political responses to the pandemic.

To overcome these challenges, this study exploits plausibly exogenous cross-county variation in lockdown exposure, operationalized as the percentage decline in mobility relative to pre-pandemic baselines (Google Mobility Reports). Counties are classified into high-exposure and low-exposure treatment groups based on the magnitude of this mobility decline. A difference-in-differences (DiD) design then estimates the causal effect by comparing the change in crowdfunding performance across these groups before and after mid-March 2020 the onset of widespread U.S. lockdowns.

Causal DAG



The DAG makes explicit that COVID Severity and County SES are open backdoor paths from Lockdown Exposure to Success. The DiD design closes these paths via county and time fixed effects. The Local Unemployment Shock node represents the potential instrumental variable pathway (green): it affects lockdown compliance through economic pressure but is assumed to have no direct channel to crowdfunding success beyond its effect on mobility.

Model Equations

Primary DiD specification:

$$Y(i, c, t) = \alpha(c) + \lambda(t) + \beta * [\text{Lockdown}(c) \times \text{Post}(t)] + \gamma * X(i, c, t) + e(i, c, t)$$

where:

- $Y(i, c, t)$ = project success (binary) or log(amount pledged) for project i in county c at time t
- $\alpha(c)$ = county fixed effects — absorb time-invariant county differences
- $\lambda(t)$ = time fixed effects — absorb nationwide shocks
- $\text{Lockdown}(c) \times \text{Post}(t)$ = treatment interaction: high-exposure county after March 2020
- $X(i, c, t)$ = project-level controls (goal, category, duration) and county controls (unemployment, income)
- β = causal ATT under parallel trends

Assumptions and Threats to Identification

Parallel Trends

High- and low-exposure counties must have followed the same crowdfunding trend before lockdowns. This is the central identifying assumption and is not directly testable, but is supported by pre-trend coefficients in the event study being close to zero.

Threat: urban counties (higher lockdown exposure) may have had faster pre-existing growth in crowdfunding. Mitigation: include pre-period county-level controls (income, broadband access) and verify event study pre-trends visually and with a joint F-test.

No Anticipation

Projects should not have adjusted their campaigns in anticipation of county-specific lockdown severity. This is plausible: lockdown orders were announced with very short notice (often 24-48 hours) and varied unexpectedly by jurisdiction.

Exogeneity of Lockdown Timing

Lockdown stringency was driven by public health criteria (case rates, hospital capacity), not by pre-existing crowdfunding trends. Mitigation: direct control for county COVID case rates and death rates to absorb the health-driven component.

SUTVA (No Spillovers)

Threat: backers in locked-down counties may redirect spending to projects in open counties, contaminating the control group. Mitigation: exclude control counties geographically adjacent to treated counties and check robustness.

Robustness Checks

- Event study pre-trends: verify $\Delta(k) = 0$ for all pre-period weeks; joint F-test.
- Placebo date: re-run DiD using March 2019 as the fake treatment date — should find $\beta = 0$.
- Alternative treatment cutoffs: classify counties at 30th, 50th, and 70th percentile mobility declines and confirm stable results.
- Continuous treatment: replace binary Lockdown(c) with the raw mobility decline percentage (dose-response).
- Alternative outcomes: $\log(\text{amount pledged})$, number of backers, and funding ratio ($\text{pledged} / \text{goal}$).
- Cluster standard errors at both county and state level; compare results.